SEMI-SUPERVISED 3D MEDICAL IMAGE SEGMENTATION USING SHAPE-GUIDED DUAL CONSISTENCY LEARNING

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ABSTRACT

Popular semi-supervised image segmentation networks suffer from two problems: firstly, supervision is only performed on the last layer of the decoder, resulting in the network's weak generalization ability; secondly, the geometry shape constraints of targets are frequently disregarded in these networks, leading to poor segmentation results. To address these issues, we propose a novel shape-guided dual consistency semi-supervised learning framework for 3D medical image segmentation. The proposed framework makes two contributions. Initially, we introduce a shape constraint to learn the shape representation, which converts the difference between two networks into an unsupervised loss and lets the model learn the boundary information of targets. Additionally, we develop a deep-supervised knowledge transfer strategy that improves the generalization ability of the network without increasing extra computation costs. Experiments demonstrate that the proposed framework outperforms state-of-the-art semi-supervised methods due to the strong ability of knowledge mining on unlabeled data.

Index Terms— 3D medical image segmentation, semi-supervised learning, shape constraint, deep supervision

1. INTRODUCTION

Medical image segmentation frequently plays a key role in computer-aided diagnosis and intelligent medicine due to the great improvement in diagnostic efficiency and accuracy [1]. Thanks to a large amount of labeled data, convolution neural networks have achieved advanced performance in medical image segmentation. Nonetheless, generating accurate annotation requires professional knowledge and clinical experience, especially in 3D medical images. Considering that the collection of unlabeled data is more convenient, it is necessary to develop semi-supervised segmentation technology to reduce the dependence on fine annotated data.

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Recently, numerous semi-supervised methods for medical image segmentation have been proposed [2] [3] [4] [5] [6]. Semi-supervised learning methods aim to improve performance by mining effective hidden information from unlabeled data [5]. Although these methods have achieved decent results, they still face the following two challenges. First of all, the training schemes only add supervision at the last layer that spread the wrong guidance information layer by layer and ignores the importance of labeled data, which may increase the risk of insufficient representation learning. Especially only a small amount of labeled data and a large amount of unlabeled data are used to train a 3D convolution neural network with a huge amount of parameters and complex structures [7]. In the second place, the above methods cannot make full use of the regional geometric constraints of targets from unlabeled data, and lack of explicit modeling of the boundary distance information, hence frequently leading to insufficient targets coverage or poor boundary prediction.

To alleviate the above problems, we propose a novel shape-guided dual consistency semi-supervised learning (SDC-SSL) framework for 3D medical image segmentation. Our framework integrates two independent networks, they have the same encoder and slightly different decoders and employ the difference of prediction to capture the uncertainty. Specifically, the segmentation model is used to generate segmentation probability maps, and the regression model is applied to generate shape-aware representation. The proposed framework contains two dominant contributions.

(1) We introduce an unsupervised shape constraint strategy based on dual consistency (SDC) to explore the boundary information of the targets so that the model pays more attention to the geometrical shape in the training process, which enhances the ability of networks for predicting the shape representation of targets in low-dimension space.

(2) We rethink the deep supervision mechanism and propose a deep-supervised knowledge transfer (DKT) strategy for 3D medical image segmentation. Without introducing extra computation costs in the inference phase, auxiliary super-

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vision branches are added to the decoder, and then the intermediate layer feature representation is transmitted as knowledge within a single network.

Experimental results indicate that our framework achieves higher segmentation performance and provides better shape prediction by mining unlabeled data and efficiently utilizing labeled data.

2. RELATED WORKS

2.1. Semi-supervised Medical Image Segmentation

To reduce the burden of annotation, many semi-supervised medical image segmentation methods have been recommended in recent years. Existing semi-supervised approaches can be roughly divided into two groups. The first is consistency learning, which assumes that even if an image encounters some disturbances, the corresponding prediction result from the sample should not change. Yu et al. explored the uncertainty and encouraged the segmentation to be consistent for the same input under various perturbations [2]. Li et al. developed a deep multi-task network and imposed a geometric shape-aware constraint on the segmentation outputs [3]. Luo et al. used a dual-task network that simultaneously predicts a pixel-wise segmentation map and a geometry-aware representation of targets [4]. Zhang et al. recommended a framework that simultaneously learns region-based shape constraints and boundary-based surface mismatches [8].

Self-training is another technique that uses labeled data to train a model initially, which is then used to generate pseudo-labels for unlabeled data. Bai et al. iteratively updated the pseudo-labels and network parameters, and then used conditional random fields to refine the pseudo-labels [5]. These approaches have been proven that the introduction of unlabeled data leads to an improvement in segmentation performance, particularly when the existing training set is small. Compared with previous methods, the proposed shape-guided dual consistency learning framework focuses on exploring the geometric boundary information of targets from unlabeled data.

2.2. Knowledge Transfer Strategy

The representative work of knowledge transfer strategy is knowledge distillation, which utilizes a large teacher model to standardize the training of a student model [9] [10]. Mutual learning is a multi-model collaborative strategy, which improves model performance in the process of collaborative training by employing the divergence and complementarity between two models [11]. Qiao et al. presented a deep co-training scheme for semi-supervised image recognition in which two efficient classifiers are established based on the complementary views of data and close predictions are assumed for unlabeled images [12]. More study implies that intermediate feature maps can be used to promote the knowledge transfer process [13].

The proposed framework contains two knowledge transfer strategies at the same time: On the one hand, the framework consists of two separate segmentation networks, which can implicitly explore useful knowledge in learning process. On the other hand, the deep-supervised knowledge transfer strategy is a high-efficiency form of knowledge transfer in a single neural network.

3. METHODOLOGY

In this section, we will introduce the proposed shape-guided dual consistency semi-supervised 3D medical image segmentation framework. As shown in **Fig.**1, our framework can be expressed as the integration of two independent segmentation networks. They share the same encoder structure and two slightly different decoders to approximate cognitive uncertainty. The segmentation network denoted by M_{seg} uses the original 3D transpose convolution to realize up-sampling, while the regression network denoted by M_{sdm} employs trilinear interpolation and 3D convolution to expand feature maps. The proposed framework contains two knowledge transfer manners. Firstly, we construct cross-task dual consistency regularization to make two networks cooperative learning with discriminative characteristics under the guidance of peer-to-peer networks. Secondly, we focus on knowledge transfer within a single network to learn more profitable features.

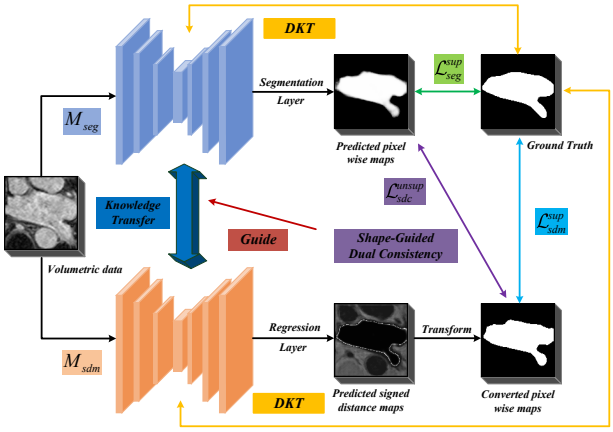

Fig. 1. The overview of the shape-guided dual consistency learning framework for semi-supervised 3D medical image segmentation. Our framework integrates two independent networks, M_{seg} is used to generate segmentation probability maps, and M_{sdm} is applied to produce shape-aware representation. In addition, auxiliary deep supervision is added to each decoder to accelerate the knowledge transfer procedure within a single network. Both networks are optimized by minimizing a combined loss including the supervised loss of labeled data, the dual consistency loss of labeled and unlabeled data, and the auxiliary loss of deep supervision.

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3.1. Shape-Guided Dual Consistency Learning

Due to the limitations of medical imaging quality, it is difficult to use a small amount of labeled data to segment targets accurately. As shown in **Fig.**2, a regression branch is added to the existing encoder-decoder structure to generate signed distance maps and capture the geometric contour information of targets [3] [14]. Then we introduce the transformation from predicted pixel-wise maps to signed distance maps [4] as follows:

$$\mathcal{T}(x) = \begin{cases} -\inf\limits_{y \in \partial S} ||x-y||_2, & x \in S_{in} \\ 0, & x \in \partial S \\ +\inf\limits_{y \in \partial S} ||x-y||_2, & x \in S_{ex} \end{cases} \quad (1)$$

where x and y are two different voxels in a predicted pixel-wise map, $||x-y||_2$ indicates the Euclidian distance between x and y , ∂S represents the contour of targets, S_{in} and S_{ex} denote the internal region and the external region of targets.

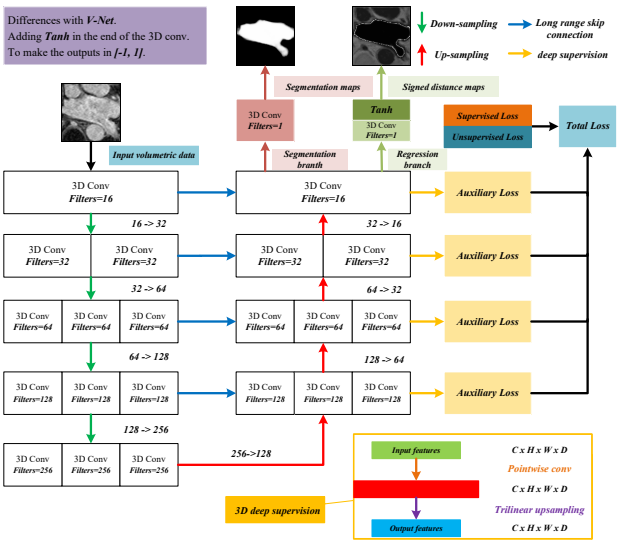

Fig. 2. Overview of the backbone for shape-guided dual consistency learning. The segmentation branch outputs the segmentation probability maps, while the regression branch outputs the signed distance maps. We append an auxiliary supervision branch after each stage of the decoder and display the structure of the 3D deep supervision module in the corner.

For the input volumetric data $x \in R^{H \times W \times D}$, the regression branch of M_{seg} is suppressed and only outputs the segmentation probability maps $y_{seg} \in [0,\ 1]^{H \times W \times D}$, while the segmentation task of M_{sdm} is restrained and only signed distance maps $y_{sdm} \in [-1, 1]^{H \times W \times D}$ are generated. In 3D medical image segmentation based on semi-supervised learning, we define the training set as $D = D_l \cup D_u$, the labeled dataset as $D_l = \{x_i, y_i\}_{i=1}^{M}$, and the unlabeled dataset as $D_u = \{x_i\}_{i=1}^{N}$, where M << N . $x_i \in R^{H \times W \times D}$ and $y_i \in \{0,1\}^{H \times W \times D}$ refers to the input image and the corresponding ground truth, respectively.

For labeled dataset D_l , each network is principally optimised by the supervised loss to learn a reliable representation of segmentation targets [8] as follows:

$$\mathcal{L}_{seg}^{sup} = \sum_{x_i, y_i \in D_l} \mathcal{L}_{dice}(f_{seg}(x_i), y_i) + \mathcal{L}_{bce}(f_{seg}(x_i), y_i) \quad (2)$$

$$\mathcal{L}_{sdm}^{sup} = \sum_{x_i, y_i \in D_l} \mathcal{L}_{dice}(\sigma(k \cdot f_{sdm}(x_i)), y_i) \quad (3)$$

where $\mathcal{L}_{seg}^{sup}$ represents the assorted loss between the predicted segmentation probability maps and the corresponding ground truth, $\mathcal{L}_{sdm}^{sup}$ indicates the compositional loss between the corresponding ground truth and pixel-wise probability maps transformed from the signed distance maps through $\sigma(k \cdot z) = \frac{1}{1+e^{-k \cdot z}}$, $\mathcal{L}_{dice}$ denotes the commonly used Dice loss, $\mathcal{L}_{bce}$ means the binary cross entropy loss. The k is a conversion coefficient as large as possible [2].

For the entire training dataset $D_l \cup D_u$, we perform an unsupervised cross-task dual consistency loss, which encourages consistent predictions on the same input images to utilize unlabeled data. In the training process, through joint learning of segmentation probability maps and geometric shape perception representations, our framework can strengthen shape constraints and collaboratively learn more discriminative features from unlabeled data. We focus on enhancing the shape-guided dual consistency between the generated segmentation probability maps and the pixel-wise segmentation maps converted from signed distance maps:

$$\mathcal{L}_{sdc}^{unsup} = \sum_{x_i \in D_l \cup D_u} \lambda_{sdc} ||f_{seg}(x_i) - \sigma(k \cdot f_{sdm}(x_i))||^2 + \mathcal{BDL}(\partial f_{seg}(x_i), \partial \sigma(k \cdot f_{sdm}(x_i))) \quad (4)$$

where λ_{sdc} indicates the rising weighting coefficient. According to the literature [4], we use a Gaussian ramp-up warming function $\lambda_{sdc}(t) = e^{-5(1-\frac{t}{t_{max}})^2}$ to control the balance between the supervised loss and the shape-guided dual consistency loss, where t denotes the present step of iterations and t_{max} indicates the maximum training step. $\mathcal{BDL}(\cdot)$ means the boundary distance loss on the geometric space, which evaluates the change between two nearby boundaries [15]. To a certain extent, the boundary distance loss can alleviate the problem of blurred edges in the segmentation probability maps since it takes the form of a distance metric on the space of contours instead of regions.

3.2. Deep-supervised Knowledge Transfer

As medical images commonly suffer from problems such as blur, noise, and low contrast, it is more difficult to extract discriminative features from medical images than other natural

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images. In the existing training schemes of semi-supervised 3D medical image segmentation networks, supervision is regularly performed only at the last layer, which ignores the importance of the small amount of labeled data, leading to problems such as the disappearance of gradients and insufficient learning of targets representations [7].

To address this problem, we propose a deep-supervised knowledge transfer strategy as shown in **Fig.**2. Specifically, the intermediate feature representation of the network is transferred as knowledge within a single network by attaching an auxiliary supervision branch to the decoder [13]. The auxiliary supervision branch is considered as a constraint condition of each up-sampling stage in the decoder, which can inject gradients from various scales.

The bottom of **Fig.**2 shows the detailed architecture of the 3D deep supervision module, where the pointwise convolution is firstly used for the input feature maps, then the trilinear interpolation is used for up-sampling, and finally, the softmax layer is used for computing the probability map of segmentation results. For labeled dataset D_l , we estimate the difference between the intermediate feature representation and the corresponding ground truth:

$$\mathcal{L}_{dkt}^{sup} = \sum_{k=1}^{K} \mathcal{L}_{dice}(h^k(x_i), y_i) + \mathcal{L}_{bce}(h^k(x_i), y_i) \tag{5}$$

where h^k means the k-th 3D deep supervision branch. Different from the existing methods, we add auxiliary supervision branches and incorporate the obtained network intermediate feature maps into the optimization process to make full use of a small amount of labeled data, thereby improving the training efficiency and speeding up the convergence of networks.

In summary, our framework is composed of two networks, M_{seg} and M_{sdm}. They cooperate in the training process and indirectly explore more reliable information from unlabeled data, but only need M_{seg} in the inference stage. The total loss of M_{seg} and M_{sdm} are defined as follows:

$$\mathcal{L}_{seg}^{total} = \mathcal{L}_{seg}^{sup} + \mathcal{L}_{sdc}^{unsup} + \mathcal{L}_{dkt}^{sup} \tag{6}$$

$$\mathcal{L}_{sdm}^{total} = \mathcal{L}_{sdm}^{sup} + \mathcal{L}_{sdc}^{unsup} + \mathcal{L}_{dkt}^{sup} \tag{7}$$

4. EXPERIMENT AND RESULTS

4.1. Dataset and Experimental Setup

We evaluated the proposed method on the public LA dataset [16] from the 2018 Atrial Segmentation Challenge. It consists of 100 3D gadolinium-enhanced MR imaging scans (GE-MRIs) and the ground truth segmentation masks, with an isotropic resolution of $0.625 \times 0.625 \times 0.625\mathrm{mm}^3$. The experiment applied the same pre-processing methods following the literature [2].

We implemented the proposed framework with PyTorch and trained it on a GeForce RTX 3090Ti GPU with 24GB VRAM. We used V-Net [17] as the backbone for all experiments, and implemented dual consistency networks by adding a regression layer at the end of the original V-Net. The framework is trained for 6000 iterations by an SGD optimizer, with an initial learning rate of 0.01 decayed by 0.1 every 2500 iterations. The batch size is 4, consisting of two labeled images and two unlabeled images. We randomly cropped $112 \times 112 \times 80$ sub-volume as the input images. According to the literature [2], we employed the standard on-the-fly data augmentation during the training stage and the value of k is set to 1500. Four frequently-used metrics are adopted to quantitatively evaluate the segmentation results, i.e. Dice similarity coefficient (Dice), Jaccard coefficient (Jaccard), Average surface distance (ASD) and 95% Hausdorff Distance (95HD).

4.2. Ablation Studies

As illustrated in **Table** 1, we conducted ablation experiments on the LA dataset using 16 labeled scans and 64 unlabeled scans to examine the effectiveness of each component. Compared with the fully supervised V-Net [17] (1) using only 16 labeled scans as the training set, the semi-supervised baseline (2) with consistency loss from the literature [4] [8] has better performance. It is demonstrated that the performance can be improved by mining meaningful hidden information from the 64 unlabeled scans when the training set is small.

To begin with, we engaged an additional shape constraint to learn the geometric representation of targets inspired by the previous semi-supervised consistency learning methods [4] [8], which facilitating the model to focus on the boundary information during training and indirectly exploring more robust knowledge from the unlabeled data. Experimental results (4)(5)(6) reveal that this technology can significantly improve segmentation accuracy. Furthermore, experimental findings (3)(5)(6) illustrate that the deep-supervised knowledge transfer approach yields significant performance gains by adding auxiliary supervised branches which assist the network in learning more discriminative and meaningful features from the limited labeled data.

Table 1. Ablation studies of the proposed framework.

Method	Loss			Metrics			
	$\mathcal{L}_{con}^{unsup}$	$\mathcal{L}_{sdc}^{unsup}$	$\mathcal{L}_{dkt}^{sup}$	Dice(%)	Jaccard(%)	ASD(voxel)	95HD(voxel)
(1)V-Net +				86.03	76.06	3.51	14.26
(2)V-Net +	✓			89.79	81.64	1.83	6.95
(3)V-Net +			✓	90.18	82.22	**1.73**	6.77
(4)V-Net +		✓		90.22	82.31	1.85	6.47
(5)V-Net +	✓		✓	90.31	82.43	1.78	6.61
(6)V-Net +		✓	✓	**90.39**	**82.57**	1.79	**6.21**

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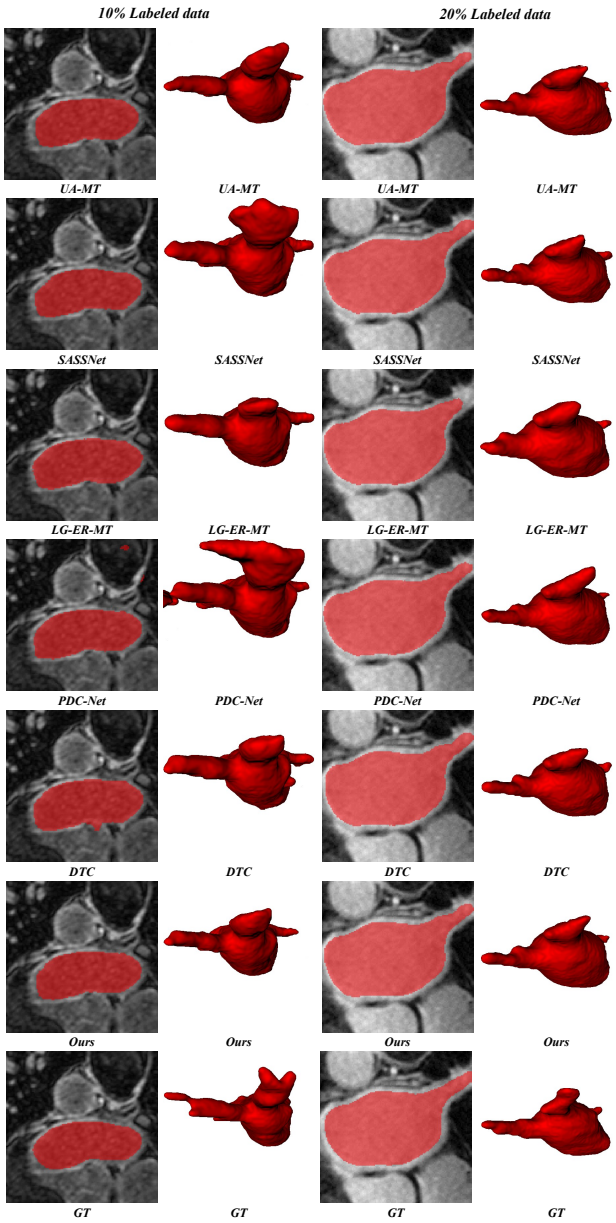

Fig. 3. Two segmentation results are obtained by UA-MT [2], SASSNet [3], LG-ER-MT [18], PDC-Net [6], DTC [4], the proposed method, and the corresponding ground truths on the LA dataset [16]. The comparisons using 10% or 20% labeled data are shown on the left or right two columns, respectively.

4.3. Comparison Experiments

In order to verify the superiority of the proposed framework, we conducted a comprehensive comparison with recent semi-supervised methods in two different settings, including DAP [19], UA-MT [2], SASSNet [3], LG-ER-MT [18], DUWM [20], PDC-Net [6], DTC [4]. Note that we used the same backbone V-Net in these methods for a fair comparison.

Table 2 shows the quantitative performance comparison of various semi-supervised methods on the LA dataset. Moreover, we specified the performance achieved by V-Net [17] as the upper and lower bounds of the task under the fully supervised setting. By effectively using unlabeled data in the training process, the proposed framework obtains considerable performance gains (from 86.03% to 90.39% of Dice). At the same time, our method achieves the equivalent results compared to the upper bound performance (90.39% vs. 91.14% of Dice), which outperforms the mainstream semi-supervised methods in other metrics. In addition, **Fig.**3 shows the visualization results of different semi-supervised methods, it is obvious that our framework produces more accurate and complete segmentation results.

Table 2. Comparisons with the state-of-the-art methods.

Method	**Scans used**		**Metrics**			
	Labeled	Unlabeled	Dice(%)	Jaccard(%)	ASD(voxel)	95HD(voxel)
V-Net [17]	8	0	79.99	68.12	5.48	21.11
V-Net [17]	16	0	86.03	76.06	3.51	14.26
V-Net [17]	80	0	91.14	83.82	1.52	5.75
DAP [19]	8	72	81.89	71.23	3.80	15.81
UA-MT [2]	8	72	84.25	73.48	3.36	13.48
SASSNet [3]	8	72	87.32	77.72	2.55	9.62
LG-ER-MT [18]	8	72	85.54	75.12	3.77	13.29
DUWM [20]	8	72	85.91	75.75	3.31	12.67
PDC-Net [6]	8	72	86.55	76.57	3.92	13.61
DTC [4]	8	72	87.51	78.17	2.36	8.23
SDC-SSL(Ours)	**8**	**72**	**88.24**	**79.13**	**1.91**	**7.86**
DAP [19]	16	64	87.89	78.72	2.74	9.29
UA-MT [2]	16	64	88.88	80.21	2.26	7.32
SASSNet [3]	16	64	89.54	81.24	2.20	8.24
LG-ER-MT [18]	16	64	89.62	81.31	2.06	7.16
DUWM [20]	16	64	89.65	81.35	2.03	7.04
PDC-Net [6]	16	64	89.76	81.57	2.95	10.31
DTC [4]	16	64	89.42	80.98	2.10	7.32
SDC-SSL(Ours)	**16**	**64**	**90.39**	**82.57**	**1.79**	**6.21**

5. CONCLUSION

This work presents a novel and practical shape-guided dual consistency semi-supervised learning framework for 3D medical image segmentation to reduce the human effort of delineating medical data. The framework integrates two independent networks that collaborate with each other to explore more reliable information from a large amount of unlabeled data, engaging an additional shape constraint to guide the learning of boundary information. We evolved a deep-supervised knowledge transfer strategy to leverage the small amount of labeled data, which upgrades the ability to capture compact features in challenging regions. Extensive experiments on public benchmark show that the proposed framework can provide a general and effective solution to achieve high-quality segmentation results.

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